4

Labor Informality and the Incentive Effects of Social Protection Systems: Evidence from a Health Reform in Uruguay

Marcelo Bérgolo
Guillermo Cruces

4.1. Introduction

Social protection systems have evolved heterogeneously in Latin America, but all the countries in the region have both contributory and non-contributory programs (Bertranou 2004).[1] For historical reasons tied to the emergence of labor movements by industrial sector, most countries have established some variant of a contributory social security. Individuals who did not have access to the system's benefits through wage contributions received some level of non-contributory social services.

Such systems can function adequately for people with coverage, and they were created with the implicit objective of achieving wide coverage. However, the countries in the region are characterized by

This chapter was financed as a part of the Regional Dialogue on Social Security organized by the Labor Markets Unit of the Inter-American Development Bank (IDB). It also received partial funding from CEDLAS' IDRC-sponsored project Labour Markets for Inclusive Growth in Latin America (www.Labor-AL.org). We wish to thank María Laura Alzúa, David Kaplan, and Carmen Pagés for comments on preliminary versions of this chapter, and attendants to the 6th IZA-World Bank Employment and Development conference in Mexico DF (May 2011). The usual disclaimer applies. The views expressed here are those of the authors and do not necessarily reflect those of their institutions or those of the funding agencies.

high levels of informality, which we define in this chapter as jobs that do not provide access to social security benefits. Given the size of the informal sector in so many countries, in the best of cases the result was an incomplete or two-tiered protection system.[2]

There is an extensive literature on the economic mechanisms explaining formal-informal labor market (see, for example, Fields 1990; Harris and Todaro 1970; Maloney 1999). Some recent studies suggest that the coexistence of partial contributory systems, universal and means-tested benefits generate a complex incentive structure, which could at least partly explain the levels of informality seen in region (Levy 2008). In a world where workers evaluate the package of benefits provided by the social protection system (with some space for the valuation of future benefits – see Summers, 1989) and the cost in terms of contributions, while employers impose the "tax" implied by the contributions with the probability of being detected and punished, workers' access to similar social benefits through the contributory and non-contributory systems would reduce the incentives for formal sector employment.[3]

This mechanism is particularly relevant in Latin America, where many countries have expanded or are planning to expand their social protection systems through (conditional and unconditional) income transfer programs and other non-contributory benefits (ECLAC 2006; Rawlings 2004). In this framework, a deep debate is needed on the consequences of the expansion of non-contributory social welfare programs for the Latin American economies, especially in terms of the effects on the interaction of the formal and informal segments of the labor market.

This study contributes to the discussion by analyzing the potential effects on informality of recent extension of health coverage to the dependants of formal workers (i.e., to be registered and contribute to the social security system) in Uruguay as part of a 2008 policy reform. This country has one of the oldest and most developed contribution-based social security systems in Latin America, with the highest coverage among the cluster of countries in the region with relatively low levels of labor informality (Gasparini and Tornarolli 2009). The system is based on employer and employee contributions from formal jobs, and provides the covered workforce with a bundle of benefits, including health coverage, and pensions among others. The 2008 exogenous expansion of the health coverage to dependent children of formal workers provides a policy experiment setting to study the

presence and quantitative relevance of the incentives of social security on labor market formality. This study's main outcome of interest is thus the worker's decision to operate formally or informally.

The rest of the chapter is organized as follows. The next section briefly describes the social protection system in Uruguay. Section 3 discusses the recent health reform and the resulting incentives on workers' decisions about working in the formal sector. Section 4 presents empirical evidence on the effects of the health reform on labor market informality. Finally, Section 5 builds on the empirical results to conduct a discussion of the policy issues that emerge from the overlap of contributory and non contributory programs in the context of informality in the labor market.

4.2. The Social Protection System in Uruguay

The social protection system in Uruguay is structured mainly around a set of contributory programs.[4] The most important institution responsible for social security benefits for the workforce is the public Social Security Bank (*Banco de Previsión Social*, or BPS), which administers services for registered salaried workers employed in the private sector.[5]

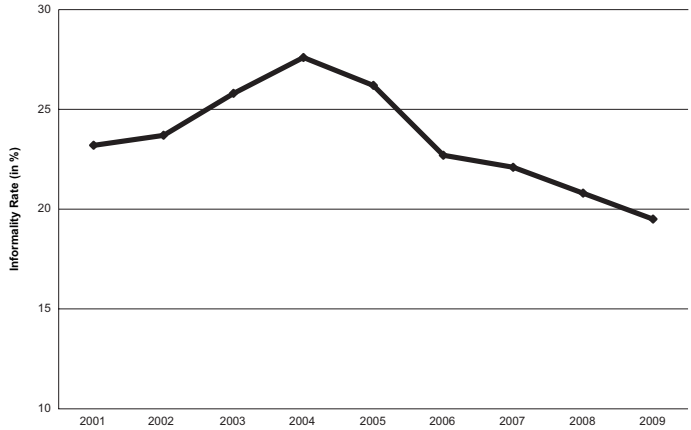
Private sector employees and employers are required by law to make contributions to the BPS, which in turn offers workers a set of services – a bundle of social benefits that is essentially provided for salaried workers in the private sector.[6] These benefits have traditionally included post-retirement benefits (retirement and pensions), as well as other benefits during the working stage of the life cycle, including access to certain healthcare services (health insurance), unemployment insurance, and family allowances for workers with dependents. While the total amount of the contributions is the sum of the different components (mainly employee contributions for health insurance and retirement pensions), the bundle of social benefits is indivisible and is tied to a single contribution that encompasses all these benefits. Consequently, BPS affiliation guarantees the worker access to all the services in the social benefits package.

Workers who are not covered by social security benefits – that is, who are not registered with the BPS or similar institutions – have access to some non-contributory programs, including non-contributory pensions and, more recently, family allowances. These programs

generally have stricter requirements, such as being below a specified income threshold. Figure 4.1 illustrates the evolution of the informality rate for salaried workers in the 2001–09 period.

Figure 4.1

Informality in Uruguay, 2001–09



Note: Informal workers are defined as salaried workers who do not have the right to a retirement pension. The informality rate represents the share of adults in informal jobs.

Source: Authors' construction, based on SEDLAC (CEDLAS and World Bank, 2011)

4.3. The Health Reform and its Labor Market Incentive Effects

4.3.1. The 2008 Health Reform

The BPS historically granted healthcare packages to registered private employees only. Under this contributory scheme, employers and workers paid 5 and 3 percent of taxable wages, respectively, for a total contribution of 8 percent. Contributing workers is eligible to select a health care provider in the private sector, paid for by the BPS. Uninsured individuals can choose to pay for their own private healthcare package, use the public healthcare system (subject to a means test), or remain uncovered.

In December of 2007 Uruguay's Parliament approved a bill (number 18, 211/2007) to reform the healthcare component of the social protection system. This bill created an integrated national healthcare system

(NHS) geared to extending comprehensive care to all residents and guaranteeing equitable and universal coverage through the coordination of the public and private healthcare sectors.[7] The pivotal component of the reform was the extension of healthcare coverage to workers' dependents, mainly children and spouses or partners. Because of fiscal restrictions, however, coverage was extended in stages. The first stage began in January 2008, when the NHS incorporated workers' children. To qualify, children must be under 18 years old. Their inclusion is universal and irrespective of previous health coverage arrangements.

The reform also substantially modified the financing of healthcare services. The new scheme is financed through a public fund called the National Health Fund (*Fondo Nacional de Salud,* FONASA; Law 18, 131/2007), which is managed by the BPS. One noteworthy feature of the reform was the increase in almost all contributions. For example, employee contributions grew from 3 to 6 percent of taxable earnings for individuals with children, while contributions for individuals with no children increased from 3.0 to 4.5 percent. Employer contributions remained unchanged at 5 percent under the new scheme.

The expansion of healthcare coverage to workers' children following the reform affects both public employees and private sector workers registered with the BPS. However, many public workers were already entitled to this extended coverage by the state, so the extension of coverage was most significant for registered private sector workers, who had to pay for their children's healthcare directly prior the reform. The new fund, FONASA, assumes this cost under the revised legislation. Even children of formal low-income workers, who previously used public health services, are now eligible for private care since FONASA entitles them to choose an IAMC in the private health sector.

4.3.2. Health Insurance Expansion and Incentives for Formal Sector Employment

The reform could potentially have a series of effects on worker incentives and subsequent labor market outcomes. Given the non-negligible level of informality in the Uruguayan labor market, the decision to operate formally or informally – that is, as a registered or unregistered worker – should be subject to an important margin of behavioral change.[8] This is especially relevant in cases of workers at the margin of the labor market and for those at the margin between the formal and informal sectors (Maloney 1999; 2004; Heckman and

Pagés 2004; Fields 2005; Gunther and Launov 2012). The decision on formal versus informal employment should be guided by a cost-benefit analysis in which individuals compare the gains from contributing to the BPS, which consist of the bundle of social security benefits described above, with its cost in terms of increased contributions.

If workers value mandated benefits – mainly those immediately available, such as health insurance – then their behavioral labor supply responses must internalize that these are only obtainable through formal employment contributions. In particular, health insurance is a substantial component of the mandated benefits package in Uruguay in terms of both the BPS's total expenditure and the quality and quantity of healthcare services provided. The expansion of healthcare coverage to other household members may thus modify the incentives for workers to operate formally. Some workers might decide to move into formal jobs or to negotiate different employment conditions with their employers in order to profit from the expansion, despite the higher cost in terms of contributions (which increased from 3 to 6 percent of taxable labor earnings for those taking advantage of this option). However, the net impact of the healthcare reform on formality is ambiguous, as it depends on whether the allocation of workers to the formal and informal sectors of the labor market is decided at the individual or household level.

From the perspective of an individual worker, the incentive to become a formal employee stems from the benefit of no longer needing to pay directly for a child's healthcare once it is covered by the social security contribution. These incentives are substantial: for a worker with an average salary, the direct cost of paying for IAMC coverage for their children is comparable to the total payroll tax of a registered worker, which provides the worker access to the whole bundle of social security benefits (including retirement savings and unemployment insurance).[9] Thus, at the individual level, the reform would potentially increase the incentives of private sector workers to become formal, and this incentive should also result in an increase in the number of children affiliated with the IAMC.

If the decision to operate formally or informally is taken jointly at the household level, the healthcare reform may have a differential impact for adults within the household depending on the intrahousehold allocation of employment relations before the policy change. Galiani and Weinschelbaum (2011) analyze this point from a theoretical perspective and also provide empirical results for Latin

America. Their study indicates that secondary workers have a higher probability of operating informally when primary workers are formal. Under the healthcare reform in Uruguay, if more than one member of the household is formal, the health coverage for children incurs a double contribution, since the law stipulates that payroll taxes are computed at the individual and not the household level. In these cases, the reform may induce some household workers to move toward the informal sector, as children would still receive coverage with only one formal worker in the household. When all workers in the household are informal, the reform may induce some of them to operate formally in order to obtain health coverage for children and spouses. Finally, in households who have members in both sectors, the policy might not induce any changes in their formality status. The incentives also remain unchanged for households with older or no children – in fact, there may even be incentives to leave the formal sector due to the increase in the payroll tax linked to the health insurance component, which rose from 3.0 to 4.5 percent for those without children.

4.4. Data and Identification Strategy

4.4.1. Data

The empirical analysis in this chapter is based on repeated cross sections of household survey microdata from the *Encuesta Continua de Hogares* (ECH) for the years 2001 through 2009. The survey is a nationwide cross-sectional household survey, which is carried out by the Uruguayan National Institute of Statistics (*Instituto Nacional de Estadística*, INE). The ECH is the main source for information on income, employment and socio-economic characteristics of households and individuals

Since 2001, the ECH includes a standard question, which asks respondents whether or not their current job entitles them to a pension or retirement savings. This question allows for quantification of the proportion of workers who are registered in the social security system.[10] Thus, this question is generally used as a proxy to identify a worker's formality status according to the legal definition (see Gasparini and Tornarolli 2009, for a discussion of alternative definitions in Latin America). In what follows, this constitutes the main dependent variable used for the econometric estimates.

The sample analysis is limited to private sector workers ages 19 through 60, who resided in urban areas.[11] The treatment group (those exposed to the policy change) consists of all adult private sector salaried workers who reside in a household with at least one child younger than 18 years old. The control group consists of adult salaried workers in the private sector with no children. The final sample is a multi-year pool of ECH microdata, including the following years: 2001-2007 and 2009 (2008, defined as an adjustment period, has been omitted).

4.4.2. Identification Strategy

The aim of the estimates presented below is to identify the causal effect of the HR on formality levels, and thus provide evidence on the direction and magnitude of the net incentive effects discussed in the previous section.[12] The empirical strategy exploits the exogenous extension of health care coverage for workers' children due to the HR in 2008 to isolate the impact of this policy on individual formality choices.[13] The econometric approach is based on the difference in differences (DD) methodology (Angrist and Krueger 1999; Bertrand, Duflo and Mullainathan 2004).

This framework compares the results of a treatment group (workers who had at least one child younger than 18) exposed to the policy change with another similar group, which feasibly represents the unobserved counterfactual evolution for the treated group (workers with no children). The estimation strategy compares formality levels among workers with children younger than 18 years and workers with no children before and after the policy change. Finally, the time period used to obtain the estimates is 2001-2009, with 2001-2007 defined as the pre-policy period and 2009 as the post-policy period, while 2008 is used as the period to capture the behavioral adjustments of workers.

4.5. Empirical Results: Incentive Effects and the Health Reform in Uruguay

4.5.1. Effects at the Individual Level

This section provides empirical evidence on how the incentives from the social security system affect labor market formality in Uruguay and the extent to which workers respond to the policy incentives

discussed above when they are making decisions about formal and informal status. Table 4.1 and 4.2 show the results discussed in the next paragraphs.

Results in Table 4.1 show the healthcare reform significantly induced private sector salaried workers with at least one child (the group affected by the reform) to switch to formal employment, with a change in the formality rate of about 1.1 and 1.3 percentage points, on average, relative to those without children (the group that was not affected by the reform). In terms of the pre-intervention average, this effect represents a 5 percent decrease in the probability to operate informally.

Table 4.2 shows heterogeneous effects of healthcare reform on labor informality for different socioeconomic groups, defined by gender, age, educational level, and firm size. In terms of gender, the effect for men is not significant and very close to zero (less than 0.3 percentage points). Most of the effects arise from the reform's impact on female salaried workers – the effect is around 2.6 percentage points, and it is statistically significant at the 1 percent level (column 3). This represents a decrease of 7 percent from the pre-policy average for women. These results are qualitatively similar to those found by Juárez (2009) and Bosch and Campos-Vázquez (2010) for the Mexican programs *Programa de Servicios Médicos y Medicamentos Gratuitos* (PSMMG) and *Seguro Popular*, respectively.

Table 4.2 also reports the estimated effect of the healthcare reform by firm size (columns 4–7). The results demonstrate negative and statistically significant effects at the usual levels for those employed in small firms (one employee) and large firms (more than 50 employees).[14] Among salaried workers employed in small firms, the reform produced a 4.6 percentage point decrease in the likelihood of working without contributing to social security, compared with a 0.8 percentage point decrease for those employed in larger firms. These findings are consistent with the fact that in smaller firms, workers may have more room to negotiate employment conditions with their employers.

Although not reported in Table 4.2, the impact of the healthcare reform also varies by age group and educational level (see Bérgolo and Cruces 2011). While the effect of the reform is negative and statistically significant across all age groups considered in the analysis, the impact is stronger for older workers than for younger workers both in absolute terms and relative to the pre-policy average informality

Table 4.1

Effect of Health Reform on Informality

	Sample: salaried workers in private sector	
	(1)	(2)
Effect of Health Reform	-0.0119***	-0.0125***
on informality	[0.0037]	[0.0033]
Socio-economic Covariates	No	Yes
Observations	84109	84109
R^2	0,04	0,31
Percent Informal (average 2001-2007)	26,33	

Note: Data are from survey years 2001-2007 and 2009 of the *Encuesta Continua de Hogares* (ECH). The dependant variable is informal status, which is equals 1 if the worker is not entitled to retirement savings. The "treatment indicator" equals 1 if the worker has at least one child and the observation is in the post intervention period, the year 2009. Controls include age, age-squared, gender, head of household status, marital status, years of education, years of education-squared, number of children in the household, and a full set of firm size, industry, departments, time dummies and a constant. Robust standard errors in brackets are clustered by year. Regressions are weighted with ECH supplement weights.

** significant at 10%; ** significant at 5%; *** significant at 1%.*

Source: Based on Bérgolo and Cruces (2011).

Table 4.2

Heterogeneous Effect of Health Reform in Uruguay on Informal Employment Levels

	By gender		By firm size (number of employees)			
	Male	Female	[1]	[2-4]	[5-49]	[+50]
	(2)	(3)	(4)	(5)	(6)	(7)
Effect of Health Reform	-0,0024	-0.0259***	-0.0469***	0,0055	-0,0057	-0.0078**
on informality	[0.0047]	[0.0037]	[0.0097]	[0.0077]	[0.0068]	[0.0029]
Observations	44171	39938	11453	14088	29755	28813
Percent Informal (avge. 2001-2007)	20,1	33,43	69,03	47,54	18,23	4,34

Note: See Table 4.1.
Source: Based on Bérgolo and Cruces (2011).
* significant at 10%; ** significant at 5%; *** significant at 1%

rate. For the youngest group (aged 19–29), the likelihood of working without contributing to social security falls by 1.0 percentage point (a decline of 3 percent), while for the oldest group (aged 50–60) the estimated impact is a decrease of about 2.1 percentage points, which represents an 8 percent reduction. Regarding educational level, the results show a negative and statistically significant effect of healthcare reform for workers with secondary and tertiary levels of education and no significant effects on workers in the low educational group. For those with secondary education, labor informality decreases by 2.0 percentage points after the reform, a 0.6 percentage point reduction rate for the tertiary education group. In both groups, this impact represents a decrease of 8 percent relative to the pre-policy period average.

4.5.2. The Impact of Health Reform on Intra-household Labor Arrangements

As discussed in Section 3, the direction of the reform's effect on formality probably depends on the existing intra-household allocation of employment relations prior to the reform. Thus, the healthcare reform may generate changes in labor arrangements within the household, since health coverage for children only requires one adult to be a formal employee. The incentive will be greatest for nuclear households in which both adults operate informally, while for other types of household the decision depends on the valuation of additional social security benefits. To account for these movements in status within the household, the study explores the possible effects of the healthcare reform on labor informality, taking into consideration the effects of potential joint decision-making by household members.

To test for this possibility, a second set of estimates uses the household, as opposed to the individual, as unit of observation. This analysis uses the same difference in difference structure as before, although, the dependent variable is defined at the household level. Column (1) explores the possible reduction in the probability for households with children in which both spouses have informal jobs. The estimates indicate a decrease in the proportion of households with children in which both spouses are informally employed. The HR results in a 4-percentage point decrease in the probability of both spouses in the same household having an informal job. Column (2) presents a second specification which tests for a re-allocation of households

from those in which both spouses were in formal employment to a household in which only one spouse was formally employed. While the sign of the coefficient of interest is negative, the estimate is small and not statistically significant at the standard levels. This result suggests that the possible incentives of the HR on a household in which both spouses are formal workers to become a household with one formal and one informal worker may be attenuated by the unwillingness of the household in the face of one member loosing other social security benefits.

As an additional check of the predictions discussed above, Column (3) of Table 4.3 compares households with both spouses operating in the informal sector with those where at least one spouse is formal. In these cases, there should be a (relative) decrease of the first type of households in response to the HR. The estimates confirm this prediction, with a 2 percentage points decrease in the likelihood of both spouses to be informally employed for households with children following the HR. This represents a 20 percent decline relative to the pre-period rates.

In sum, the results indicate that the healthcare reform had a sizable impact on women, middle-aged and older workers, and workers with secondary and higher education. In addition, the reform's effect seemed to be more relevant for workers employed in small firms. This evidence suggests that the reform increased the incentives for workers to operate formally, either by renegotiating their employment conditions with their employers or by moving to other formal jobs. The evidence also indicated that household members react jointly to the change in the incentive structure implied by the extension of benefits to children.

For the case of Uruguay, these results support the expected theoretical prediction that improving the benefits from a mandated social package provides additional incentives to become a formal worker. They also imply a positive valuation of mandated social security benefits for a subset of the population.

4.6. Conclusions and Policy Discussion

This chapter analyzed the impact of the introduction of a health insurance reform on labor informality in Uruguay – the extension of health coverage to the children of private sector salaried workers

Table 4.3

Effect of Health Reform on Formality Joint Decision of Households. Estimation Results for Dependent Variable

	Dependent variable: formality status of spouses in the household		
	1= Both informal 0 = Exactly one spouse formal (1)	1 = Exactly one informal 0 = Both spouses formal (2)	1 = Both informal 0 = At least one spouse formal (3)
Effect of Health Reform	-0.0421***	-0,0058	-0.0198***
on informality	[0.0130]	[0.0057]	[0.0041]
Observations	4812	13466	14592
R^2	0,20	0,29	0,22
Percent both Informal (avge. 2001-2007)	26,16		9,75
Percent one Informal (avge. 2001-2007)		30,49	

Note: See Table 4.1.
* significant at 10%; ** significant at 5%; *** significant at 1%.
Source: Based on Bérgolo and Cruces (2011).

which contribute to the social security system. The results indicate that the HR had a sizable impact on women, middle age and older workers, and for workers with secondary and higher education. In addition, the effect of the HR seemed to be more relevant for workers employed in small firms. This evidence suggests larger incentives for workers to become formal in jobs in which they have greater bargaining power, re-negotiating employment conditions with their employers or moving to other formal jobs. The evidence also indicated that household members react jointly to the change in the incentive structure implied by the extension of benefits to children.

The empirical results support the expected theoretical prediction that improving the benefits from a mandated social package provides additional incentives to become a formal worker. Furthermore, these results seem to be robust to different specifications and controlling for additional dimensions. The main finding is that labor informality decreased in Uruguay due to the HR's expansion of health care coverage, and workers and households were able to react to the new incentive structure. The results also imply a positive valuation of social security mandated benefits for a subset of the population.

Evidence that social security systems imply sizeable incentive effects for workers, and that workers react to these incentives, has serious implications for the design of labor market policies and social protection systems in developing countries. This is especially relevant in the context of the expansion of non-contributory elements such as conditional cash transfer programs. Like most countries in Latin America, Uruguay has expanded its social protection system in the past decade. One thing that Uruguay did differently, however, was to base some of the expansion on its contributory programs – the pensions system, health insurance, and unemployment insurance. Furthermore, the extension of the income transfer system to groups that have traditionally been excluded from coverage was accomplished within the existing traditional social welfare framework – namely, through the expansion of the family allowance program. Thus, instead of extending differentiated benefits and services to people who could not make contributions, the new scheme (which was mainly implemented in the second half of the 2000s) represented an original alternative to the new regional paradigms in the area of social welfare (see ECLAC 2006, chap. 5).

The extent to which the changes to these programs are able to satisfy coverage objectives while at the same time improving the willing-

ness of workers to contribute to social security depends, in part, on the workers' valuation of the costs and benefits of social security, the alternatives to these costs and benefits, the magnitude of the changes introduced, and the degree of consistency among the incentives they generate. In some cases, the design of these reforms appears to be consistent with better incentives for workers to contribute to social security and thus to obtain coverage through the bundle of social security benefits. In other cases, the changes may have improved worker coverage at the expense of generating adverse incentives to registering for social security. In still other cases, the changes introduced to the programs have probably generated ambiguous incentives.

The empirical evidence on the effect of these reforms on formality is scarce. With regard to the extension of health insurance to cover the children of registered workers, the evidence suggests that the reform improved workers' willingness to contribute to social security. In principle, the expansion of this contributory program appears to have reconciled the objectives of improving population coverage and providing incentives for workers to operate formally. The pensions system reform, however, does not appear to have achieved the same result. In this case, we do not find evidence that the penalty in terms of the rate of return for short contribution histories, after the reform in 1995, has changed the contributions profile of workers. Moreover, some studies indicate that a very large share of workers will not reach the minimum years of contributions required to retire from the labor market at the normal retirement age, even after the requirement was lowered from 35 to 30 years in the 2008 reform (Bucheli, Forteza, and Rossi 2008).

Some of the reforms appear to have narrowed the coverage gaps, but the gaps are still large, in particular those associated with the contributory social protection system (Arim, Cruces, and Vigorito 2009). The strategy of expanding the contributory programs to narrow the coverage gaps and at the same time improve the levels of formality in the economy clearly has limitations. One possible restriction has to do with the difference between the State's and the worker's valuation of contributory benefits and how these benefits are provided to the workers – bundling benefits. The state requires workers to buy a whole set of goods and services, so as to fulfill certain objectives that are considered socially necessary. This bundling requires workers to assess a single set of components that have widely different characteristics – present versus future consumption, collective versus individual needs, monetary versus in-kind receipts,

everyday use versus emergencies, and so forth. The imperfect match between the individual needs (or preferences) of workers and the set of benefits in the bundle implies that workers may not want to consume all the benefits, or at least not all at the same time. The impossibility of splitting up the package of benefits lowers the value of social security for workers and thus reduces their willingness to contribute (Levy 2008). After the healthcare reform in Uruguay, many workers may be willing to contribute 6% of their wages to obtain health insurance coverage for their families, but they do not want to contribute additional resources to gain access to the rest of the benefit bundle. While the expansion of health insurance has increased the formality incentives for a group of workers, the share of individuals with health coverage would probably be greater if they could make contributions for this benefit alone.

In addition, there is a limit to how far coverage can be expanded through the redesign of contributory social welfare programs, given by the current productive and labor market conditions of a developing country like Uruguay. One indication of this restriction is the difficulty of lowering the levels of informality even in periods of strong growth, such as the 2004–10 period in Uruguay.

Given this framework, what are the options for continuing to expand the social protection system in Uruguay? One option is to continue expanding and/or reforming the contributory social welfare scheme so as to generate incentives for workers to make contributions. Given the limitations discussed above, this strategy requires rethinking the criteria for supplying the bundle of social benefits – for example, considering the possibility of breaking up the bundle and having workers voluntarily "buy" the type and quantity of social goods and services desired based on their individual preferences. The workers' valuation will be influenced by the available alternatives and their costs, as well as by the quality of these alternatives (non-contributory benefits and services provided by the market). An alternative that does not alter the compulsory nature or indivisibility of the bundle of social benefits, while still being based on the expansion of the contributory system, would require deep changes in the design of the bundle so as to improve the incentives to contribute or at least eliminate mechanisms that generate counter-incentives. One possibility would be to consider the household, rather than the individual, as the basis for contributions. In Uruguay, as in most countries, each worker must pay for all the benefits in the bundle even when some

of the benefits extend to the family, thereby generating situations of overlapping contributions by different family members. The health reform provides an example of this design issue: in nuclear households, both spouses must make contributions to extend their health coverage to their children, but one of the spouses does not "enjoy" the additional benefit implied by the extra contribution (that is, only one contribution is necessary to cover the children). This incentive may become more important in 2011, when health coverage is extended to the spouses of formal workers (as established by the Law).

A second option would be to expand the existing non-contributory programs or to implement new programs that are not tied to the contribution history of the beneficiaries, so as to provide coverage to workers and their families who participate in the labor market without contributing to social security. In this case, the scale would be tipped toward coverage rather than incentives. For example, the extension of the family allowance program en Uruguay has huge benefits, but it undoubtedly removes one of the distinctions between formal and informal employment: under the new system, everyone receives the allowance, but only some "pay" for the benefits implicitly through their contributions. The same situation arises with less immediate benefits like retirement, where less stringent qualifications for the benefits are currently under discussion, or even the expansion of the non-contributory component of pensions. The evidence on the incentive mechanisms affecting formality and informality in the region's labor markets suggests that the coexistence of contributory and non-contributory programs – while providing broader coverage of the population – probably undermines the incentives to contribute to social security.

Finally, at the other extreme, there is the possibility of moving toward a universal system financed through taxes (that is, universal incomes for the elderly and children, universal access to unemployment and health insurance, and so forth). Under certain regulatory conditions, this could be considered the "first best" option in terms of equity, efficiency, and coverage (see Levy 2008). However, these universal social protection systems appear to be difficult to implement in the short and medium terms, since there are so many different interests involved and so many norms and regulations that would need to be modified.

The near future for Uruguay and the rest of the countries in Latin America will probably be sealed by a "second best" scenario, in which

the traditional systems live side by side with ever-growing non-contributory programs. The key may lie in conducting a cost-benefit analysis of the programs and then defending those that appear to have a positive balance in net terms.[15] In this case, one potential problem is that the presence of disincentives leads to the rejection of measures with a potential positive net impact.

In the absence of governments with a broad mandate to reform social security and to institute far-reaching transformations, it will be necessary to design policy options that minimize the incentive conflicts (in terms of informal work, labor market participation, and so forth). This is ultimately the objective of the research agenda that is being developed in the region on the interaction between social protection systems and labor markets.

Notes

1 Contributory programs are those in which financing and access to benefits (such as healthcare, old-age pensions, and workers' compensation) are directly related to contributions made by the workers, their employers, and, in some cases, the state. In contrast, non-contributory programs are not tied to the workers' contribution history, and their financing is largely dependent on taxes (Bertranou 2004).
2 See Gasparini and Tornarolli (2009) for a thorough analysis of the evidence for the region.
3 Levy (2008) considers contributions and premiums to be taxes on formality when there is also a non-contributory benefits system.
4 Although, in the past there were also programs that did not depend on the beneficiary's contribution history (Bertranou, 2005; Ferreira-Coimbra and Forteza 2004).
5 The other institutions give coverage to other groups of workers. In particular, two social security institutions cover to the police and armed forces, respectively. The other three institutions administer services for professionals and for workers in the financial sector. Each of these institutions provides different bundles of benefits, usually broader than those provided to BPS beneficiaries.
6 These benefits are generally also provided to public sector employees when they sign an employment contract.
7 In implementing the healthcare reform, the government sought to strengthen three areas: healthcare coverage (focusing on primary care); health management; and health financing.
8 Whether workers have the power to make that decision or whether it all comes down to an employer's cost-benefit analysis is a contentious issue in the literature. Instead of opting for one of the two extremes, we assume that the worker has some degree of influence in this decision (see for instance, Levy 2008).
9 In 2008, the average monthly wage for salaried workers with children was around US$630 (at 2005 purchasing power parity), of which about US$130 corresponded to social security contributions for formal workers. In the same period, the average amount that workers paid to enroll a child in an IAMC was US$60. Since salaried workers have two young children, on average, the total amount paid for private healthcare would amount to US$120.
10 In addition, starting in 2008 after the implementation of the HR, the ECH also includes a question about whether or not individuals are entitled to health insurance by the NHS.
11 See the working paper version of this chapter (Bérgolo and Cruces, 2011) for details about the sample restrictions and summary statistics for treatment and control groups in the pre-policy and post-policy periods.
12 For more details on the empirical strategy see Bérgolo and Cruces (2011).
13 Boyle and Lahey (2010) and Gruber and Madrian (1995) among others, use a similar approach to analyze the effect of health coverage extensions on different outcomes in the US labor market.
14 The information recorded by the ECH does not support a further breakdown of wage earners working in firms of "more than 50 employees".
15 Cruces and Gasparini (2008) discuss these aspects in the context of the universal family allowance in Argentina.